

CalibAgent: Task-Aware Active Calibration and Shift Recovery for Quadruped Velocity Commands

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Abstract—Commercial and learned quadruped controllers often expose a planar velocity command while hiding the dynamics that map commands to realized motion. Payload, surface, controller, and estimator changes can therefore preserve balance yet distort navigation. We present CALIBAGENT, an active calibration layer for this opaque interface. A Bayesian command-to-motion model represents cross-axis coupling and asymmetric response; a task-weighted integrated variance objective selects informative commands for the intended deployment distribution. Hard safety checks, validation-based stopping, constrained inverse compensation, and residual-triggered recovery remain separate from the learned acquisition score. On 183 passive Unitree Go2 trials, a coupled affine model reduces held-out velocity RMSE by 54.5% relative to raw commands. In registered synthetic experiments, task weighting reaches joint accuracy and uncertainty targets in 18.67 trials, saving 39.5% against Latin hypercube sampling and 18.8% against D-optimal design. Pinned Isaac Lab experiments show 9.3–31.7% held-out RMSE reduction, faster early recovery under four shifts, and successful budgeted navigation across six prospectively frozen maps. These results establish the calibration and recovery mechanism under the registered conditions; active hardware calibration and navigation are reserved for the planned real-robot study.

I. INTRODUCTION

A quadruped can remain upright while executing the wrong motion. High-level controllers accept a desired planar velocity $\mathbf{u} = [v_x, v_y, \omega_z]^\top$, but the steady body velocity may contain gain error, dead zones, saturation, and cross-axis coupling. These errors are particularly consequential when a planner assumes that the exposed command is a calibrated control variable. Taouil *et al.* showed that learned motion models of a black-box quadruped controller can improve footstep planning [1]; related work has calibrated velocity-controlled kinematic models online for wheeled visual-inertial odometry [2]. The remaining question is how to collect the calibration trials themselves when hardware time is limited and the useful part of command space is task dependent.

Robot calibration has long treated measurement configuration as a design variable [3], [4]. Classical observability and D-optimal criteria prioritize parameter identification [5], [6]. Task-oriented calibration instead scores a design by uncertainty at the end-effector or another quantity of interest [7], [8]. We apply that principle to the external velocity interface of a fixed quadruped controller: trials are chosen for their reduction of predictive uncertainty over commands expected during deployment, not for uniform parameter-space coverage.

This layer differs from locomotion-policy adaptation. RMA conditions a trained policy on latent environment es-

timates [9], residual models can update model-based legged controllers online [10], and recent embodiment adaptation identifies hardware changes from short interaction histories [11]. CALIBAGENT neither retrain nor instruments the low-level controller. It identifies the mapping from user commands to measured steady motion, translates desired velocities through a constrained inverse, and reactivates calibration when predictive residuals indicate a shift.

This paper makes four contributions. First, it formulates the opaque command-to-motion interface as a compact Bayesian regression with coupled, asymmetric features and per-trial measurement uncertainty. Second, it derives an exact task-weighted integrated variance reduction (IVR) score for sequential command selection. Third, it separates statistical learning from hard safety, validated stopping, and bounded shift recovery. Fourth, it evaluates the chain from model need to downstream navigation using passive Go2 data, controlled synthetic families, fault injection, pinned physics simulation, and a retained failed confirmation followed by a disjoint prospective replication. The current hardware evidence is passive; real online calibration, shift recovery, and navigation remain outside the reported results.

II. RELATED WORK

A. Calibration and Experimental Design

Kinematic and dynamic robot calibration depend on both model structure and the poses or trajectories used for identification [3], [4]. Sun and Hollerbach linked robot observability indices to alphabetic optimal design and developed an efficient active pose-selection algorithm [5], [6]. Carrillo *et al.* then selected configurations by predicted task-space error rather than only parameter covariance [7]. This task-space view parallels goal-oriented Bayesian design, which minimizes uncertainty in a downstream quantity of interest [8]. Greedy information selection also has established connections to near-optimal sensor placement [12]. CALIBAGENT uses a related predictive objective, but its design variable is a safety-filtered velocity command and its quantity of interest is the realized velocity over a frozen deployment distribution.

B. Legged Command Models and Adaptation

Simulation-trained policies have produced agile and terrain-robust quadruped locomotion [13], [14], [15]. Robust locomotion, however, does not make the user-level velocity interface exact. Learned black-box motion models can bridge that interface for planning [1], while steady-state and local dynamics models can adapt a legged robot to unmodeled disturbances [16]. Policy-level adaptation and

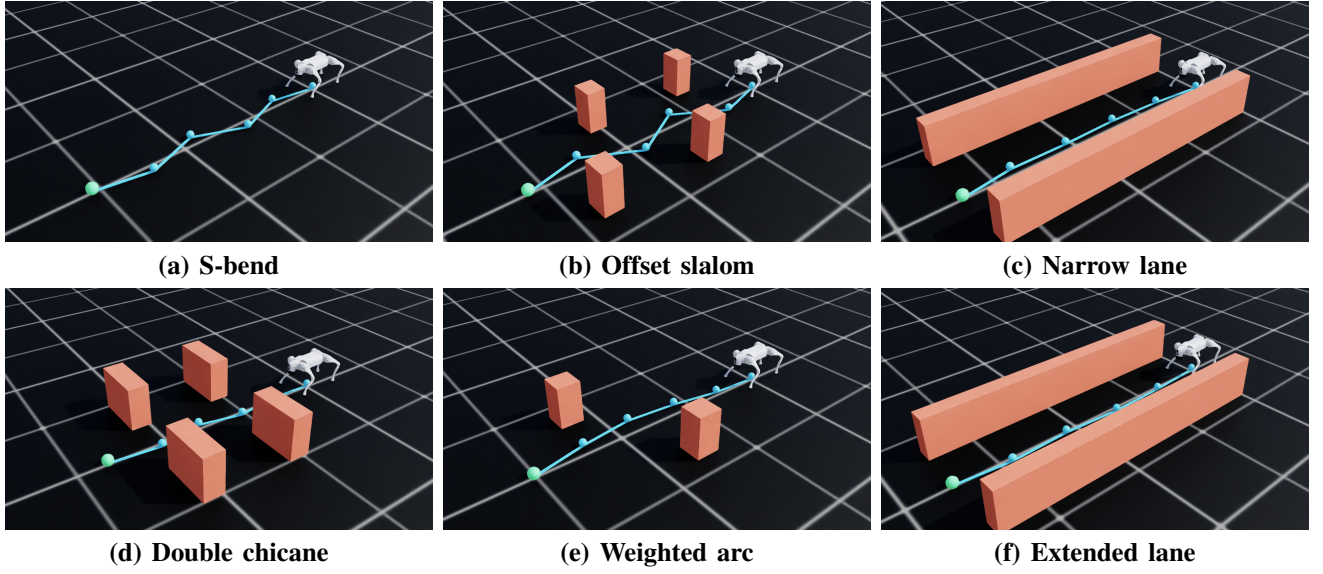


Fig. 1. Elevated Isaac Sim views of all six frozen P7 prospective-replication maps. Cyan segments and spheres reproduce the map waypoints; orange geometry reproduces the registered obstacles. The overlays are non-colliding visualization geometry and do not contribute to the quantitative endpoints.

domain randomization address broader changes [9], [17], [11]. CALIBAGENT occupies a narrower operational layer: it actively calibrates a fixed controller’s command response, quantifies posterior uncertainty, and applies compensation only inside a non-learned envelope.

III. METHOD

Figure 2 separates the probabilistic calibration loop from the rules that constrain its operation. The low-level locomotion controller is fixed and treated as a black box. A trial executes a constant body-frame command after warm-up and ramp-in, extracts a robust steady-motion observation, and records its covariance and validity. Invalid observations never update the posterior.

A. Bayesian Command-to-Motion Model

Let $\mathbf{y} \in \mathbb{R}^3$ denote the measured steady velocity for command $\mathbf{u}$. A fixed basis $\phi(\mathbf{u})$ contains an intercept, the three command axes, pairwise products, and positive/negative hinge terms for each axis. The affine M1 variant uses only the intercept and command axes; M2 adds the coupled and hinge features. Feature statistics are fitted once on a predeclared command reference, never on held-out outcomes.

For output axis a , the model is

$$y_a = \theta_a^\top \phi(\mathbf{u}) + \epsilon_a, \quad \epsilon_a \sim \mathcal{N}(0, \sigma_a^2 + r_a), \quad (1)$$

where r_a is the observation covariance diagonal. Given prior $\mathcal{N}(\mathbf{m}_{a0}, \Sigma_{a0})$, sequential updates use

$$\Lambda_a \leftarrow \Lambda_a + \frac{\phi\phi^\top}{\sigma_a^2 + r_a}, \quad (2)$$

$$\eta_a \leftarrow \eta_a + \frac{\phi y_a}{\sigma_a^2 + r_a}, \quad \Sigma_a = \Lambda_a^{-1}, \quad \mathbf{m}_a = \Sigma_a \eta_a. \quad (3)$$

The predictive mean is $\mu_a(\mathbf{u}) = \mathbf{m}_a^\top \phi(\mathbf{u})$, and the predictive variance is $\phi^\top \Sigma_a \phi + \sigma_a^2$. Independent output posteriors

keep the online update small while coupled input features represent cross-axis response.

B. Task-Weighted Active Selection

A frozen task distribution is represented by grid commands $\{\mathbf{g}_j\}_{j=1}^G$ and normalized weights w_j . For candidate $\mathbf{c}$, the posterior covariance update is independent of the unknown outcome. The exact reduction in weighted epistemic variance is therefore

$$\mathcal{I}(\mathbf{c}) = \sum_{a=1}^3 \sum_{j=1}^G w_j \frac{(\phi(\mathbf{g}_j)^\top \Sigma_a \phi(\mathbf{c}))^2}{\sigma_a^2 + \phi(\mathbf{c})^\top \Sigma_a \phi(\mathbf{c})}. \quad (4)$$

The planner maximizes $\mathcal{I}(\mathbf{c}) - \lambda_r R(\mathbf{c}) - \lambda_d D(\mathbf{c})$ over a finite command pool after excluding commands too close to prior trials. Greedy batches use covariance-only fantasy updates, so no unobserved target is imputed. The registered main planner uses 512 candidates, an eight-command seed design, a 400-point task grid, and a 0.025 normalized duplicate radius.

C. Safety, Stopping, Compensation, and Shift Recovery

The hard filter checks finite state, localization, battery, roll, pitch, base height, command bounds, linear norm, coupled linear-angular load, command slew, and projected workspace occupancy. It evaluates ranked candidates after planning and fails closed when none remains. A 50 Hz monitor can zero the command independently of the planner. This separation prevents uncertainty reduction from trading against a learned safety penalty.

Stopping requires minimum trial coverage and held-out validation. A run stops at the registered target only after three consecutive confirmations of both validation RMSE and integrated uncertainty; a low-gain condition also requires valid held-out accuracy. Trial, time, distance, and battery

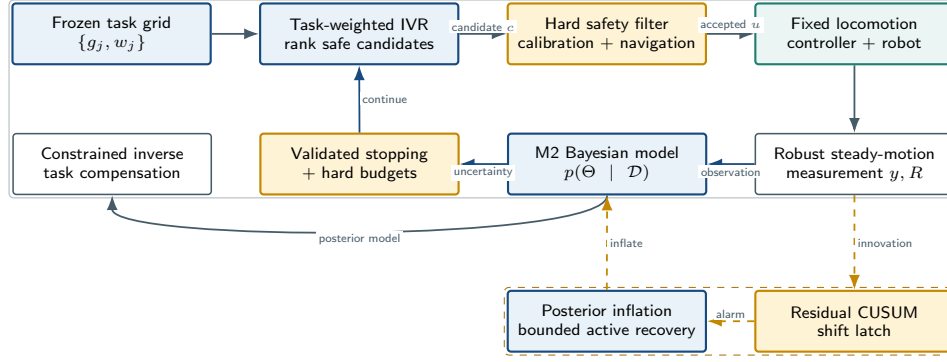


Fig. 2. CALIBAGENT architecture. Task-weighted acquisition ranks calibration commands, but the hard filter alone authorizes execution. Valid observations update the Bayesian model; held-out validation and uncertainty control stopping. A latched residual alarm inflates posterior uncertainty and opens a bounded active-recovery phase. The inverse compensator uses the same bounded candidate set during navigation.

budgets take priority. For deployment, the inverse compensator searches the same bounded candidate set and minimizes

$$\|\mu(\mathbf{u}) - \mathbf{v}^*\|_2^2 + \lambda \|\mathbf{u} - \mathbf{v}^*\|_2^2 + \rho \sum_a s_a^2(\mathbf{u}), \quad (5)$$

then passes the selected command through the hard filter.

Shift detection uses normalized innovation energy $z_t = \mathbf{e}_t^\top \mathbf{S}_t^{-1} \mathbf{e}_t / 3$. A one-sided CUSUM accumulates $q_t = \max(0, q_{t-1} + z_t - 1 - \kappa)$ and alarms only after a frozen threshold, minimum dwell, and repeated positive evidence in a bounded window. The alarm latches, inflates epistemic covariance without changing the posterior mean, and starts a capped recovery budget. Normal stopping resumes after recovery.

IV. EXPERIMENTAL PROTOCOL

The evaluation follows a staged evidence ladder. P1 contains 183 valid passive Unitree Go2 trials from three 61-trial sessions. Livox MID360 FAST-LIO/global localization provides the reference pose. Each complete session is held out in turn; training budgets are 30 trials for sampled designs and 122 for the dense reference. P1 tests model structure, not online active execution.

P3 supplies the controlled sample-efficiency test. Twenty independent seeds span affine, dead-zone, and heteroscedastic families. Family conditions are averaged within seed, so the inferential sample size is 20 rather than 60. Controls are random, Latin hypercube (LHS), Sobol, Bayesian D-optimal, active-without-task, and a 256-command dense oracle. The target jointly requires held-out RMSE and epistemic uncertainty. Paired effects use 4,000-sample seed bootstrap intervals and one-sided paired Wilcoxon tests.

P4 replays the frozen active trajectories through the stopping rule and injects proposal and runtime faults. P5–P7 run in Isaac Lab v2.3.2 at commit 37ddf62 with Isaac Sim 5.1.0-rc.19 and PhysX [18]. P5 contains four scenarios with 20 paired seeds, 12 active trials, and eight validation commands; Fig. 3 shows the four registered simulator contexts. P6 contains four held-out in-place shifts, 72 paired

seeds per shift, and frozen, passive-update, and full active-recovery controls (Fig. 4). P7 contains six disjoint maps, seven calibration methods, and 72 new paired seeds per map: 3,024 navigation episodes. Figure 1 shows all six registered maps. Dense uses 30 trials; LHS, Sobol, D-optimal, no-task, and full use 12. Every method shares the frozen waypoint follower and map-specific planner hash. Rate endpoints use exact binomial intervals; continuous paired endpoints use 4,000-sample bootstrap intervals.

V. RESULTS

A. Model Need and Calibration Efficiency

Figure 5a shows a consistent held-out benefit from coupled calibration on the passive Go2 data. Pooled RMSE falls from 0.06605 for raw commands to 0.04563 for diagonal M0 and 0.03009 for coupled M1. M1 therefore reduces error by 54.45% versus raw commands and 34.07% versus M0. The three held-out-session M1 values are 0.02888, 0.03158, and 0.02974. The 30-trial LHS M1 result is 3.33% above the 122-trial dense reference (0.02912). These data support a coupled correction for this robot and acquisition day; they do not measure active hardware efficiency.

P3 isolates the acquisition rule (Fig. 5b). The full method reaches the joint target in 18.67 trials, compared with 30.87 for LHS, 30.67 for random, 25.72 for Sobol, 23.00 for D-optimal, and 25.67 without task weighting. Reductions are 39.52%, 39.13%, 27.41%, 18.84%, and 27.27%, respectively; all one-sided paired Wilcoxon tests give $p = 9.54 \times 10^{-7}$. The primary LHS comparison saves 12.20 trials with a 95% interval of [10.73, 13.73]. Final active RMSE is 0.008462, 6.90% above the dense oracle’s 0.007916, and 95% predictive coverage is 94.14%.

In closed-loop P5 simulation (Fig. 3e), raw-to-calibrated RMSE is 0.15199 to 0.10635 for affine distortion, 0.15054 to 0.10282 for dead-zone distortion, 0.10741 to 0.09742 for low friction plus payload, and 0.09554 to 0.07849 on rough terrain. Relative reductions span 9.30–31.70%. The corresponding paired absolute-improvement intervals are [0.04290, 0.04780], [0.04495, 0.05002], [0.00981, 0.01017],

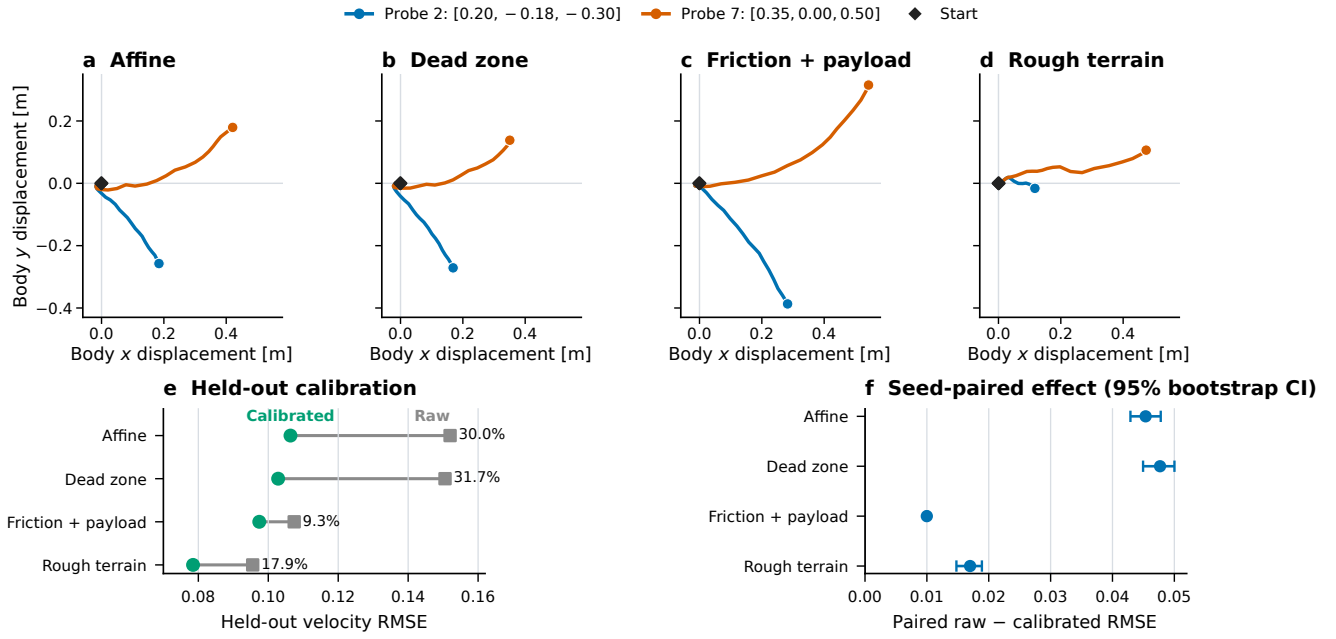


Fig. 3. P5 registered simulation responses and closed-loop calibration. (a)–(d) Body-displacement trajectories for two validation probes at display seed 5301; diamonds and circles denote the common start and measured endpoints. (e) Raw and calibrated held-out velocity RMSE across 20 paired seeds; labels give relative reduction. (f) Seed-paired absolute improvement with 95% 4,000-sample bootstrap intervals. The trajectories are qualitative examples; all inference uses the registered paired summaries.

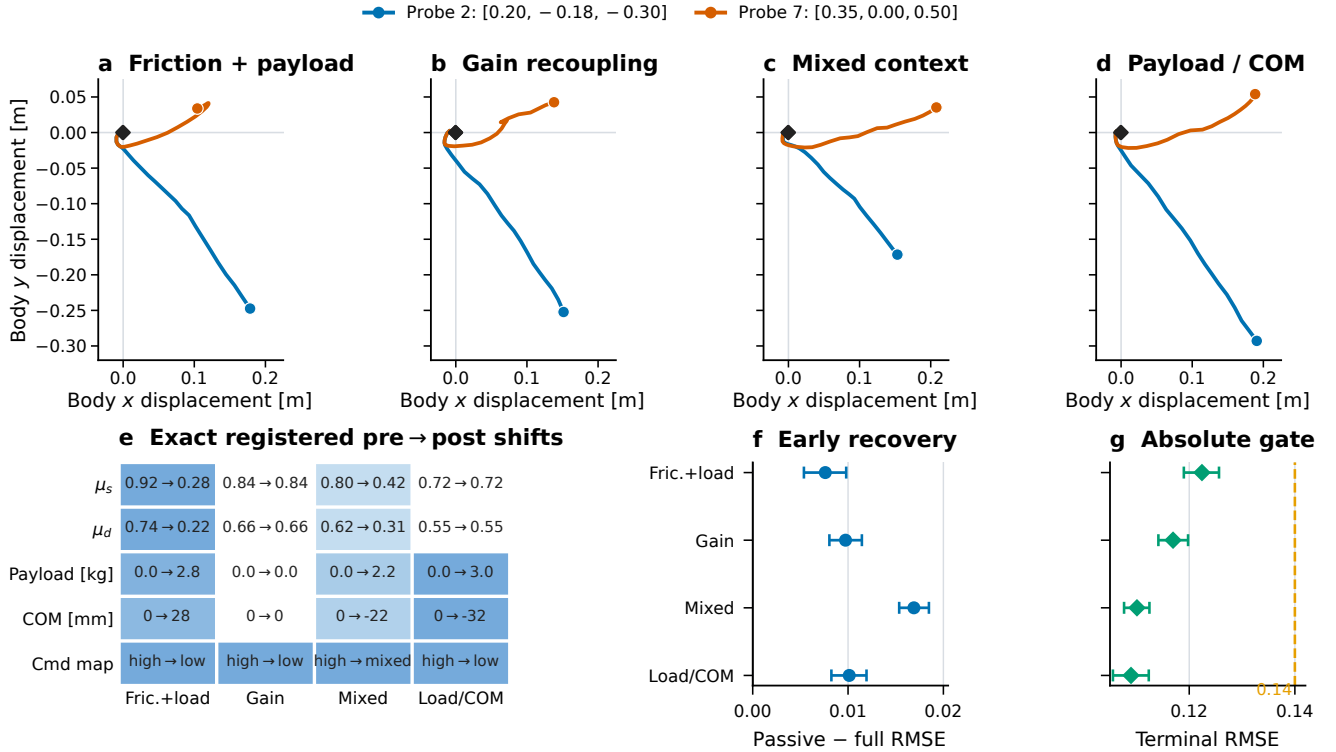


Fig. 4. P6 registered held-out shifts and active recovery. (a)–(d) Body-displacement trajectories for the same validation probes at display seed 10101. (e) Exact pre→post static and dynamic friction, payload, fore-aft COM, and command-map settings; color intensity encodes within-row shift magnitude. (f) Full-method improvement over passive updating in the registered early window. (g) Full-method terminal RMSE and the frozen 0.14 gate. Error bars are paired 95% 4,000-sample bootstrap intervals over 72 seeds per shift.

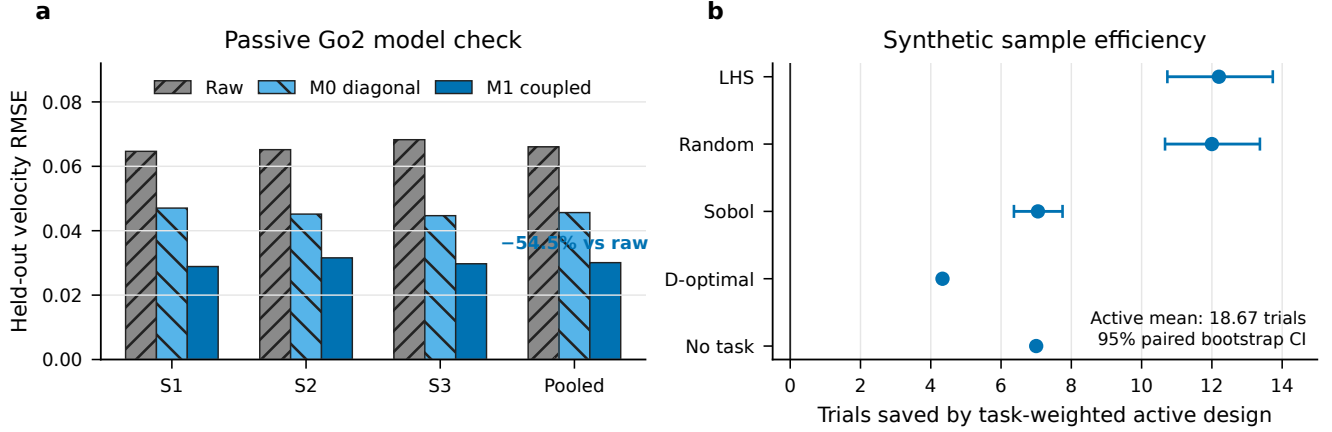


Fig. 5. Calibration evidence. (a) Passive real-Go2 held-out RMSE for raw, diagonal M0, and coupled M1 commands. (b) Paired trials saved by the task-weighted active method; bars are 95% seed-bootstrap intervals for the paired difference.

and [0.01477, 0.01889]. All 20 paired seeds improve in each scenario.

B. Stopping, Safety, and Shift Recovery

P4 records no premature stop across 60 trajectories. Median and p95 overshoot after the oracle target are both two trials. The hard filter rejects 300/300 registered hazards and 0/20 safe controls; the runtime monitor catches all 160 fault injections with a maximum recorded detection latency of 0 ms. These are software replay and fault-injection results, not physical safety certification.

Across the four P6 shifts, the full method improves early RMSE over passive updating by 0.00761 for friction plus payload, 0.00974 for gain recoupling, 0.01691 for mixed context, and 0.01012 for payload/COM (Fig. 4f). Their 95% intervals are [0.00537, 0.00981], [0.00804, 0.01145], [0.01536, 0.01848], and [0.00825, 0.01194]. Detection succeeds in 72/72 seeds for three shifts and 71/72 for payload/COM; full recovery succeeds in 72/72 for every shift. Worst p95 detection and recovery delays are four and six trials. Full terminal RMSE ranges from 0.10898 to 0.12241, with every upper interval endpoint below 0.126 (Fig. 4g). Passive updating can be competitive at the terminal window, so the supported contrast is the registered early-recovery effect.

C. Navigation Consequence and Replication

The first P7 strong confirmation did not pass. Full-method success was 68/72 on one map and 50/72 on another, and some completion-time ratio interval upper bounds reached 1.347. Trace inspection exposed a 10 Hz guard blind interval in which base height could cross the hard limit between planner ticks. A 50 Hz predictive height interlock was developed, frozen, and evaluated on six new maps and seeds. No result from the failed confirmation or intervening pilots is pooled with the replication.

In the disjoint replication, B8 succeeds in 72/72 episodes on five maps and 70/72 on weighted arc (Fig. 6a); every B8

collision count is zero. Paired B8-versus-raw completion-time gains have lower interval endpoints from 23.453 to 32.675 s (Fig. 6b). The worst B8/dense mean time-ratio interval upper bound is 1.0736, below the registered 1.25 margin, and the worst ratio upper bound against any matched-budget control is 1.0901 (Fig. 6c). Because planner hashes and seeds are matched, the comparison isolates command calibration within the frozen simulator pipeline. It does not establish hardware navigation or sim-to-real transfer.

VI. DISCUSSION AND LIMITATIONS

The experiments separate three questions that are often conflated. P1 asks whether the exposed command interface needs coupled calibration. P3 asks whether task weighting changes which trials are valuable. P7 asks whether a calibration that looks accurate on held-out commands also preserves navigation outcomes. The no-task and D-optimal controls show that the P3 gain is not merely generic active learning, while the P7 protocol prevents validation RMSE from rescuing a failed downstream endpoint.

The evidence has a clear hardware boundary. P1 is passive, same-day data from one Go2 in one environment. P5–P7 use a pinned simulator and fixed policy. The compact steady-response basis does not represent arbitrary history dependence, gait switching, rapid transients, or shifts outside the registered families. The planned P8 study must therefore execute active calibration on hardware, apply gain/coupling, payload/COM, friction, and mixed shifts, and test navigation on weighted-arc and offset-slalom courses. Until those data pass their frozen gates, the present result is a simulator-validated calibration and recovery architecture supported by passive hardware model evidence.

VII. CONCLUSION

CALIBAGENT calibrates the high-level velocity interface of a fixed quadruped controller by directing trials toward task-relevant predictive uncertainty. The approach combines a coupled Bayesian model, exact IVR acquisition, hard safety

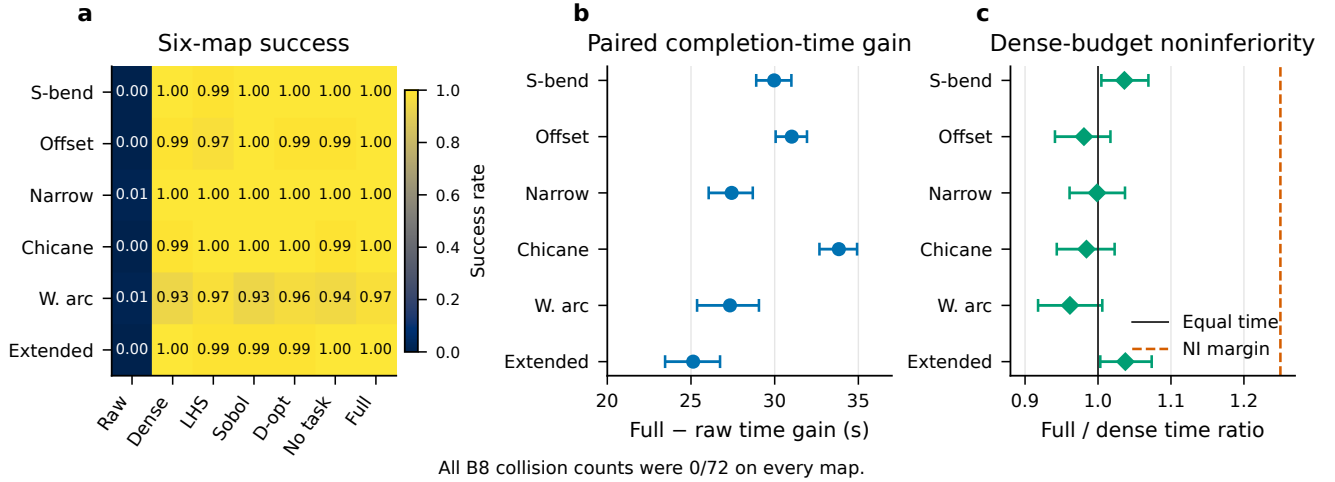


Fig. 6. P7 prospective simulator replication. (a) Success rate for six maps and seven methods (72 seeds/cell). (b) Paired completion-time gain of full calibration over raw control. (c) Full/dense time ratio with paired 95% intervals; the registered noninferiority margin is 1.25. Every B8 collision count is 0/72.

filtering, validated stopping, constrained inverse compensation, and latched shift recovery. Passive Go2 measurements establish the need for a coupled map; registered synthetic and Isaac Lab experiments establish sample efficiency, early shift recovery, and downstream navigation under controlled conditions. The remaining test is active real-robot execution of the same frozen protocol.

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